Solution Requirements

Date	2 July 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely access the application and submit credit card application details.	High
2	Authorization levels	The administrator can manage datasets, train the model, and monitor predictions. Users can only submit applications and view prediction results.	High
3	External interfaces	The system uses the uploaded CSV dataset and machine learning libraries	High

		(Scikit-learn, XGBoost) for prediction.	
4	Transactions processing	The system accepts applicant details, preprocesses the data, and predicts whether the credit card application is likely to be approved or rejected.	High
5	Reporting	Displays the prediction result to the user and provides model performance metrics such as accuracy.	Medium
6	Business rules, etc.	Credit card approval is predicted based on applicant information such as income, employment status, credit history, and other relevant attributes.	High
7	Compliance to laws or regulations	User data should be handled securely and confidentially in accordance with applicable data privacy guidelines.	Medium
8	*Other*	The system should allow easy retraining of the machine learning model using updated datasets.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should generate prediction results quickly.	Prediction displayed within 2 seconds.
2	Scalability	The application should support multiple users and larger datasets without significant performance degradation.	Supports 100+ users and larger datasets efficiently.
3	Security & Data Privacy	Applicant information should be securely stored and protected from unauthorized access.	Secure handling of user data with controlled access.
4	Reliability & Availability	The application should provide consistent prediction results and remain available during normal operation.	99% availability during usage.
5	Usability & Accessibility	The application should provide a simple, user-friendly interface for entering applicant information and viewing results.	Easy navigation with clear instructions and responsive design.
6	*Other*	The system should be easy to maintain, update, and retrain with new datasets and machine learning models.	The system should be easy to maintain, update, and retrain with new datasets and machine learning models.